

Automated Eligibility Criteria Extraction

Current Technical Architecture, Semantic Outputs, and MetricSpace Foundation

Modern AutoCrit Project

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1 Purpose and Current Scope

Modern AutoCrit is a local browser application that extracts clinical-trial eligibility criteria from a ClinicalTrials.gov record, pasted eligibility text, or an uploaded full protocol PDF. It produces human-reviewed, structured criteria intended to support the MetricSpace project: comparison of eligibility rules by clinical concept, logical meaning, quantitative constraint, and contextual modifiers.

The current output is deliberately richer than a flat Entity–Attribute–Value record. Each atomic criterion separates its clinical domain and entity from its comparator, value, unit, interval, negation, temporal restriction, repetition, qualifier, Boolean relationship, and verbatim source. This separation prevents phrases such as “blood pressure > 145/95 on three occasions” from collapsing into one opaque entity string.

The current application does *not* include the retired desktop interface, the deleted clinical-distance screen, the former candidate-attribute critic/reconciliation system, automatic job resumption, or a full Chia annotation graph. Model output remains a review artifact rather than clinical ground truth.

2 Lineage and Design Changes

The project began from the protocol-to-structure idea demonstrated by AutoCriteria [1, 2]. The implementation now differs in several ways:

- LangChain orchestration was replaced by a small provider interface and the OpenAI SDK.
- The original long script was divided into registry, LLM, workflow, normalization, persistence, HTTP, and browser modules.
- Desktop interfaces were replaced by a local, full-width web application.
- ClinicalTrials.gov API v2 and full-protocol PDF ingestion were added.
- Human decisions are stored in timestamped JSON and CSV snapshots instead of silently changing the immutable packaged base.
- The former EAV output was replaced by a Chia-inspired structured-criterion projection. Chia distinguishes domain entities, value fields, semantic constructs, and relations; Modern AutoCrit adopts the useful distinctions in a compact tabular form without claiming full annotation-graph compatibility. See <https://www.nature.com/articles/s41597-020-00620-0>.
- Explicit interval endpoints were added for MetricSpace. Equivalent range expressions can therefore share one computational representation.

3 User Workflow

The browser contains three panels.

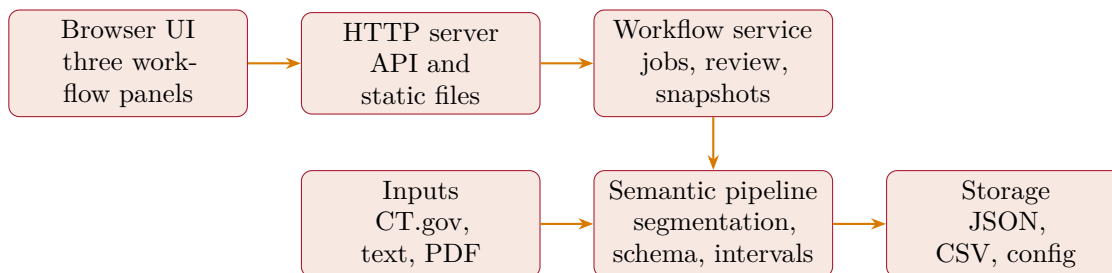
Extraction / ingestion. Select an NCT identifier, paste eligibility text, or upload a complete protocol PDF. PDF ingestion searches for the actual inclusion and exclusion sections and excludes later protocol material such as the SAP and schedule of activities.

Human review. Sort the extracted rows, accept correct rows, or reject a row to edit its domain, entity, comparison, interval endpoints, semantic constructs, and Boolean relation. A corrected row must be saved before the collection can be applied.

Criteria library. Search the immutable base criteria and the latest reviewed snapshot. Historical EAV snapshots are upgraded in memory for display; their original files are not rewritten.

During extraction, the progress panel reports protocol loading, the number of located source criteria, completed extraction batches, retries, semantic normalization, completion, or failure.

4 Architecture



Component	Responsibility
<code>main.py</code>	Parses host, port, and browser options and launches the application.
<code>webapp/server.py</code>	Routes the JSON API, limits request bodies, serves static assets safely, and starts the threaded local server.
<code>webapp/static/</code>	Implements ingestion, progress polling, sortable review, inline correction, and library search.
<code>backend/services/ctgov_v2_service.py</code>	Validates NCT identifiers, retrieves API v2 study JSON, and returns eligibility text.
<code>backend/services/llm/</code>	Provides the provider-neutral boundary, OpenAI Responses API calls, strict JSON parsing, retry behavior, and token accounting.
<code>backend/eav_pipeline.py</code>	Retains its historical filename but now owns the structured-criterion schema, source segmentation, terminology normalization, domain inference, interval parsing, semantic constructs, and relation labels.
<code>backend/services/eav_workflow.py</code>	Persists jobs, extracts PDF sections, batches model calls, records progress, upgrades historical snapshots, validates review decisions, and writes criteria-library snapshots.
<code>config/terminology_normalization.json</code>	Contains clinical aliases, exact-only ambiguous abbreviations, and mappings from legacy entity classes to Chia-style domains.
<code>config/base_criterion_library.js</code>	Supplies immutable structured seed examples.
<code>runtime/criteria/</code>	Stores ignored local job records and runtime library copies.

5 Extraction and Normalization Sequence

1. The browser posts the selected input mode and its content to `/api/extractions`.
2. The workflow validates required input. NCT mode retrieves ClinicalTrials.gov; PDF mode extracts text and locates the actual eligibility section; manual mode uses the pasted text.

3. `segment_criteria()` identifies inclusion and exclusion headings, joins wrapped lines, separates numbered or bulleted criteria, splits independent semicolon-delimited cutoffs, and assigns stable source identifiers.
4. Criteria are sent in bounded batches through `StructuredJsonAdapter` under the strict structured-criterion schema.
5. The model separates coordinated clinical concepts and proposes domains, values, units, temporal phrases, repetition, qualifiers, negation, and AND/OR relationships.
6. `normalize_rows()` rejects invented source strings, restores authoritative inclusion/exclusion labels, adds missing source rows, applies deterministic literal parsing, normalizes terminology, and generates stable row identifiers.
7. The browser presents every row for explicit human acceptance or correction.
8. Applying review writes a cumulative, versioned `criteriaLibrary_*.json` and `criteriaLibrary_*.csv` to both runtime storage and `config/`.

6 Structured Criterion Model

6.1 Domains, fields, constructs, and relations

The compact model adapts four useful aspects of Chia:

- **Domain:** observation, condition, person, device, drug, visit, procedure, or measurement.
- **Entity:** the canonical clinical concept only, without its cutoff or modifier.
- **Fields and constructs:** comparator, value, unit, interval bounds, negation, temporal restriction, repetition, and qualifier.
- **Relations:** derived labels such as `HAS_VALUE`, `HAS_NEGATION`, `HAS_TEMPORAL`, `HAS_MULTIPLIER`, `HAS_QUALIFIER`, `AND`, and `OR`.

Rows from the same source retain a shared `source_id`. This preserves the relationship between atomic concepts split from one sentence.

6.2 Intervals

A numerical rule contains nullable lower and upper bounds with separate inclusivity flags. The application also emits a readable interval string. Examples include:

Source	Interval	Interpretation
<code>age > 18</code>	<code>(18, +inf)</code>	Open lower endpoint.
<code>age <= 65</code>	<code>(-inf, 65]</code>	Closed upper endpoint.
<code>aged 18 to 65</code>	<code>[18, 65]</code>	Closed bounded range in years.
<code>between 18 and 65 years</code>	<code>[18, 65]</code>	Equivalent normalized range.
<code>age >=18 and <=65 years</code>	<code>[18, 65]</code>	Equivalent explicit comparator form.

Ratios such as blood pressure 145/95 remain literal values rather than scalar intervals. This avoids pretending that a compound measurement is one scalar cutoff.

6.3 Illustrative structured output

For “Blood pressure > 145/95 on three occasions,” the output separates:

- domain: `measurement`; entity: `Blood Pressure`;

- comparator: >; value: 145/95;
- repetition: on three occasions;
- relations: HAS_VALUE and HAS_MULTIPLIER;
- verbatim source and stable source identifier.

7 Terminology Governance

The terminology configuration combines three legacy Modern AutoCrit maps with current additions. All legacy attribute and disease aliases are retained. Legacy entity labels are adapted into the current domains rather than treated as disease synonyms. For example, “lab test” maps to **measurement**, “medication” to **drug**, and “diagnosis” to **condition**.

Oncology coverage includes common diseases, performance measures, laboratory concepts, response concepts, and biomarkers, including NSCLC, SCLC, AML, CLL, DLBCL, ECOG performance status, Karnofsky performance status, RECIST, ANC, ALT, AST, ULN, EGFR, ALK, HER2, PD-1, PD-L1, KRAS, BRAF, BRCA1, and BRCA2.

Ambiguous short forms such as ALL, MM, CR, PR, SD, and PD are exact-match only. Thus an extracted entity equal to ALL can normalize to Acute Lymphoblastic Leukemia without rewriting the ordinary phrase “all patients.”

8 Function Reference

run_browser_app()

Creates the threaded server, optionally opens the browser, and closes server resources on interruption.

BrowserApplication.dispatch()

Routes health, extraction submission/status, review application, and library requests.

fetch_study() / eligibility_text_from_study()

Retrieve one ClinicalTrials.gov API v2 record and its eligibility block.

EavWorkflowService.submit() / get()

Validate input, persist a queued job, start its worker, and retrieve its durable JSON state. The historical class name remains for compatibility.

EavWorkflowService._eligibility_from_pdf()

Read a complete PDF, find the actual eligibility section, remove page furniture, and stop before later numbered sections.

EavWorkflowService._run()

Coordinate segmentation, batched model calls, retries, normalization, progress, completion, and failure capture.

EavWorkflowService.library()

Load the immutable base and latest reviewed snapshot, upgrading historical EAV rows in memory.

EavWorkflowService.apply_review()

Validate complete human review and atomically write cumulative JSON/CSV snapshots.

segment_criteria()
Create inclusion/exclusion source items and stable source identifiers.

extraction_prompt_for_items()
Build the source-bounded AutoCriteria/Chia-inspired extraction instruction.

deterministic_criterion()
Parse literal comparison and range syntax, separate semantic constructs, normalize the entity, infer a domain, and derive relation labels.

canonical_term()
Apply exact and phrase-safe aliases while protecting ambiguous abbreviations.

infer_domain()
Use governed domain aliases and deterministic clinical cues when a valid model domain is unavailable.

interval_notation()
Serialize open, closed, bounded, and unbounded numerical constraints.

semantic_relations()
Derive compact Chia-style relationship labels from the structured fields.

normalize_rows()
Reconcile model rows with authoritative sources and produce final review rows.

StructuredJsonAdapter.generate_json()
Request schema-constrained model output and parse a JSON object.

OpenAIProvider.generate()
Call the Responses API with retry/backoff and token recording.

atomic_write_json()
Write through a temporary file and atomic replacement.

9 Outputs and Their Meaning

9.1 Job record

Each submission creates `runtime/criteria/jobs/<job_id>.json`. It contains the job and trial identifiers, input mode, timestamps, status, stage, progress percentage, user-facing message, error when present, extracted rows after completion, and an ordered progress-event log.

Status	Meaning
<code>queued</code>	The durable job record exists and its worker has not started.
<code>running</code>	Ingestion, model extraction, or semantic normalization is active.
<code>completed</code>	Structured rows are available for human review; they are not yet governed additions.
<code>failed</code>	The job stopped and <code>error</code> contains the immediate exception.

9.2 Review rows

Every row contains `row_id`, `source_id`, criterion type, domain, entity, comparator, value, unit, value kind, interval endpoints and notation, negation, temporal restriction, repetition, qualifier, logical operator, derived relations, and source. Human rejection opens these fields for correction; saving the correction marks the row accepted.

9.3 Criteria-library snapshots

Applying review creates cumulative files named `criteriaLibrary_YYYYMMDD_HHMMSS_microseconds.json` and `.csv`. JSON is authoritative and includes `schema_version=modern-autocrit.structured-criteria.v3`. CSV provides a convenient tabular export. Copies are written to `runtime/criteria/libraries/` and `config/`; generated config copies are ignored by Git.

The application can read historical `eavLibrary_*.json` snapshots. It converts their comparator/value form into the richer representation in memory and does not mutate the old file. The first newly applied review carries the upgraded records into a v3 criteria snapshot.

10 Errors and Operational Interpretation

Error or symptom	Meaning and response
Missing API key	Add <code>OPENAI_API_KEY</code> to the repository-root <code>.env</code> and restart.
Invalid or missing NCT ID	Check the identifier and registry availability.
No eligibility criteria segmented	Headings or list formatting were not recognized; use manual text or inspect the PDF section.
PDF section not recognized	The document lacks recognizable inclusion/exclusion headings or is image-only. OCR is not currently performed.
Structured response failure	The provider did not return the strict schema after retries. The job is marked failed rather than returning silent partial data.
Request body too large	The server accepts at most 75 MB JSON bodies and the browser limits PDF selection to 55 MB.
Review cannot be applied	Every row must be accepted or corrected; entity, domain, value/bounds, comparator, and logic are validated.
Port already in use	Stop the prior process or launch with a different <code>-port</code> .
Legacy snapshot shown	No v3 snapshot exists yet; the newest historical EAV file is being upgraded in memory. Apply a reviewed run to create v3 files.

11 Current Limitations

- Jobs are persisted but interrupted jobs are not automatically resumed; model prompts and raw responses are not yet written as a complete audit trail.
- A failed extraction batch currently fails the job instead of preserving partial rows for continuation.
- PDF extraction is text-based and does not OCR scanned protocols.

- The criteria library stores reviewed trial assertions. A future governed concept layer should separate reusable concept definitions from trial-specific constraints.
- Terminology normalization is a curated alias system, not an ontology service. RxNorm, LOINC, OMOP, NCIt, or ICD links are not yet assigned.
- The representation is inspired by Chia but does not implement arbitrary nested annotation graphs or executable patient queries.
- MetricSpace distances, calibration, and learned representations are not yet implemented. The current schema supplies their inputs.

12 MetricSpace Roadmap

The structured output permits distinct comparison channels:

- canonical-entity agreement and governed synonymy;
- domain compatibility;
- exact interval relations, including equivalence, subset, superset, overlap, and disjointness;
- comparator direction and endpoint sensitivity;
- negation, inclusion/exclusion polarity, temporal scope, repetition, qualifier, and Boolean-structure disagreement;
- embedding or LLM judgment for residual clinical semantics that cannot be resolved structurally.

Reviewed human corrections can become labeled supervision. A future agentic layer can select structural comparison for exact constraints, retrieve terminology candidates, invoke an embedding model for paraphrases, request LLM adjudication for ambiguity, reconcile disagreement, quantify uncertainty, and escalate unresolved cases.

Near-term engineering priorities are batch checkpoints, raw model audit records, an independent critic, separation of canonical concepts from trial assertions, ontology identifiers, and a labeled criterion-pair evaluation set. A learned dense representation should be trained only after the structural baseline and human labels are reliable.

13 Conclusion

Modern AutoCrit is now a browser-delivered protocol eligibility extractor and human review tool whose output is designed for comparison rather than display alone. Its v3 schema preserves quantitative intervals, semantic constructs, Boolean relations, normalized oncology terminology, and source evidence. The application does not yet calculate the final MetricSpace distance, but it supplies a substantially more defensible representation on which that statistical and agentic work can be built.

References

- [1] S. Datta, *AutoCriteria*, GitHub repository.
<https://github.com/surdatta/autocriteria>
- [2] S. Datta et al., *AutoCriteria: A Generalizable Clinical Trial Eligibility Criteria Extraction System Powered by Large Language Models*, 2023.
<https://arxiv.org/abs/2304.06096>